Deep Order-Flow in Prediction Markets: Forecasting Short-Horizon Mid-Price Returns on Polymarket

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Abstract

We ask whether deep order-flow forecasting techniques developed on equity limit order books transfer to prediction markets, using twelve Polymarket NBA series-winner contracts (April–May 2026) as a test bed. Three architectures—a single-layer LSTM, an LSTM with a small MLP head, and a CNN-LSTM—are trained on signed, level-resolved order flow to jointly forecast mid-price returns at horizons of $1, 2, 3, 5, 10$ ticks, and are evaluated against a naive train-mean predictor and a Ridge linear benchmark on out-of-sample R^2_{OS}, directional accuracy, and an annualized Sharpe ratio. All three neural models beat both baselines on every metric, reaching roughly 61% directional accuracy on moved ticks. Two structural findings reproduce in this new venue: a plain LSTM nearly matches the larger CNN-LSTM, so the value lies in the order-flow representation rather than architectural depth; and predictability is concentrated at the nearest tick and decays quickly with the horizon. The transfer, however, is established at the level of signal rather than realized profit: the squared-error edge is fragile out of sample, reflecting the non-stationarity of contracts marching toward a fixed resolution, and the reported Sharpe ratios are frictionless.

1 Problem and Motivation

Prediction markets have grown into a meaningfully sized financial venue, yet their empirical microstructure remains understudied: existing work emphasizes aggregate forecast accuracy, calibration, and cross-contract arbitrage (Wolfers and Zitzewitz, 2004; Manski, 2006) rather than the short-horizon dynamics of the order book itself. This contrasts with the equity microstructure literature, where deep models have become standard for short-horizon mid-price forecasting from limit order book (LOB) state—DeepLOB (Zhang et al., 2019), deep order-flow (Kolm et al., 2023), and more recently TLOB (Berti et al., 2025)—with the consistent finding that deep models fed order flow (OF) rather than raw book state recover signal that linear and naive predictors miss. Since Polymarket's central limit order book is mechanically a standard CLOB, we ask whether that finding transfers from equities to prediction markets.

Concretely, for a given contract let mid_t denote the mid-price at discrete time step t. We forecast multi-horizon mid-price returns $r_{t,h} = \mathrm{mid}_{t+h} - \mathrm{mid}_t$ for horizons $h \in \mathcal{H} = \{1, 2, 3, 5, 10\}$ from the most recent 100 steps of order flow; the target is therefore a vector (an alpha term structure). We ask whether a deep model beats (i) a naive constant predictor and (ii) a Ridge regression on the same window, evaluating along three axes: statistical accuracy (R^2_{OS}), directional accuracy on non-flat ticks, and the annualized Sharpe ratio of a sign-following strategy. Positive results would be initial evidence that mature equity-LOB techniques translate to prediction markets; mixed results would be informative about where these venues differ.

2 Data and Preprocessing

We use tick-level Level-2 order-book data for Polymarket, a binary venue where each contract settles to the realized outcome of an event. Data come from the Telonex Polymarket Historical Data API,[1] whose `book_snapshot_25` channel records every order-book change (rather than sampling at fixed intervals); we retain the ten best levels per side. We download the twelve NBA playoff "series-winner" contracts traded over April–May 2026, keeping one tradeable outcome per series. These predict which team wins a best-of-seven series, and are distinct from single-game markets. After cleaning, the dataset comprises 2,041,832 snapshots, with per-market depth from $\approx$ 39,000 rows (Suns vs. Thunder) to $\approx$ 377,000 (Lakers vs. Rockets).

The mid-price $\text{mid}_t = \frac{1}{2}(\text{bid}_t^0 + \text{ask}_t^0)$ lies in $[0, 1]$ and is read as the market-implied probability of the outcome. The forecasting targets are the forward changes $r_{t,h} = \text{mid}_{t+h} - \text{mid}_t$, predicted jointly across the five horizons. Because order flow is meaningful only when the mid is well defined, we admit a contract only if its tight-spread fraction (ticks with spread below 0.05) is at least 60%; all twelve satisfy this screen (Table 1). Cleaning proceeds in four steps: snapshots are time-sorted (order flow is a first difference); the $\approx$ 7–10% exact/near-simultaneous duplicates are collapsed to the last state within each microsecond; malformed books (missing best level or inverted quote $\text{bid}_t^0 \geq \text{ask}_t^0$, 0.03–0.63% of rows) are dropped; and missing deep-level entries are imputed consistently with the order-flow construction (missing bid price 0, ask price 1, size 0).

Inputs are the three-case order-flow features of Cont et al. (2014). For a bid level i with price b_t^i and size v_t^i,

$$\text{OF}_t^{b,i} = \begin{cases} +\,v_t^i & b_t^i > b_{t-1}^i \quad \text{(new bid)} \\ v_t^i - v_{t-1}^i & b_t^i = b_{t-1}^i \quad \text{(size change)} \\ -\,v_t^i & b_t^i < b_{t-1}^i \quad \text{(bid pulled)}, \end{cases} \tag{1}$$

with the ask side symmetric. Concatenating ten bid and ten ask levels gives $\text{OF}_t \in \mathbb{R}^{20}$; a single input is the look-back window of $W = 100$ ticks, a 100×20 matrix, never crossing a market boundary. Each feature is winsorized at its $[0.5, 99.5]$ training quantiles (raw order flow spans $\approx \pm 6 \times 10^5$, so this controls outliers) and then standardized to zero mean and unit variance; targets are standardized analogously, so a constant mean predictor attains a standardized MSE of exactly one. All statistics are estimated on the training split alone, so no future information enters preprocessing. Finally, for the robustness analysis of Section 8 we partition the twelve markets into Low/Mid/High volatility tertiles by $\text{vol} = \text{std}(\text{diff}(\text{mid}))$, which captures per-tick jitter rather than the overall travel of the path.

3 Exploratory Data Analysis

Three features of the cleaned data each motivate a downstream choice.

Scale and heterogeneity. Table 1 inventories the twelve contracts. The markets are highly heterogeneous: raw book updates span a factor of nine ($\approx$ 45,000 to $\approx$ 406,000), and the number of *mid-price changes*—the events actually forecast—is far smaller and varies even more (from under 4,000 to $\approx$ 68,000). Each contract lives only six to seventeen days. These small, uneven samples are the principal motivation for a single pooled model (Section 4). The mid-price paths (Figure 1) are visibly non-stationary, with volatility clustering in bursts; this is why we split chronologically and estimate all standardization statistics per market on the training split alone.

[1] The `book_snapshot_25` channel returns event-driven, tick-by-tick snapshots of the top twenty-five levels per side in a flattened price/size schema, captured on every book update over websocket. https://telonex.io (accessed June 2026).

Table 1: **Per-market data inventory.** "Updates" is the raw book-update count before cleaning; mid changes, mean spread, and tight-spread fraction are on cleaned books. "OF turnover" is $\sum_t \sum_i (|\mathrm{OF}_t^{b,i}| + |\mathrm{OF}_t^{a,i}|)$ over the top ten levels, a book-activity proxy in share units (not executed volume).

Market	Winner	Updates	Lifetime	Mid chg.	Spr. (¢)	Tight	OF (M sh.)
Lakers vs. Rockets	Lakers	406,120	14d 15h	54,906	1.83	0.931	7,168
Timberwolves vs. Nuggets	Timberwolves	347,075	13d 16h	24,443	2.74	0.916	4,445
76ers vs. Celtics	76ers	288,151	15d 13h	68,198	1.78	0.948	6,527
Raptors vs. Cavaliers	Cavaliers	277,046	16d 12h	59,013	2.53	0.904	3,702
Pistons vs. Magic	Pistons	260,332	15d 7h	32,616	2.36	0.864	4,304
Spurs vs. Trail Blazers	Spurs	127,975	11d 15h	20,059	2.39	0.876	1,846
Knicks vs. Hawks	Knicks	126,891	13d 12h	6,590	3.16	0.862	1,284
Spurs vs. Timberwolves	Spurs	125,728	14d 13h	6,928	2.09	0.918	2,011
Cavaliers vs. Pistons	Cavaliers	114,162	13d 13h	10,461	4.16	0.695	5,746
Thunder vs. Lakers	Thunder	52,398	9d 18h	5,039	1.05	0.999	638
Knicks vs. 76ers	Knicks	49,072	6d 9h	5,482	2.07	0.930	3,651
Suns vs. Thunder	Thunder	44,636	9d 13h	3,952	1.24	0.943	400

Figure 1: **Mid-price paths over each contract's lifetime,** colored by the chronological validation/training/test split. Heterogeneous and non-stationary, motivating the per-market chronological split and standardization.

Order flow is weak, distributed, and mean-reverting. The order-flow imbalance $\mathrm{OFI}_t = \sum_i \mathrm{OF}_t^{b,i} - \sum_i \mathrm{OF}_t^{a,i}$ is strongly anti-persistent (autocorrelation ≈ -0.22 at lag one) and its predictive content is weak and mean-reverting: contemporaneous with the mid-move it correlates $\approx +0.09$, but with the *next* tick's return only ≈ -0.03, the signature of a bid–ask bounce. Crucially, this predictive content is spread across many lags with alternating sign out to ≈ 20 ticks (Figure 2), so a naive equal-weight sum over the window nearly cancels (≈ -0.007 with the next return) and is in fact worse than the single most recent tick (≈ -0.035). The signal is real but nonlinear and distributed through time—which is why the Ridge benchmark fails out of sample and a model that learns a nonlinear, recency-aware aggregation is the right tool.

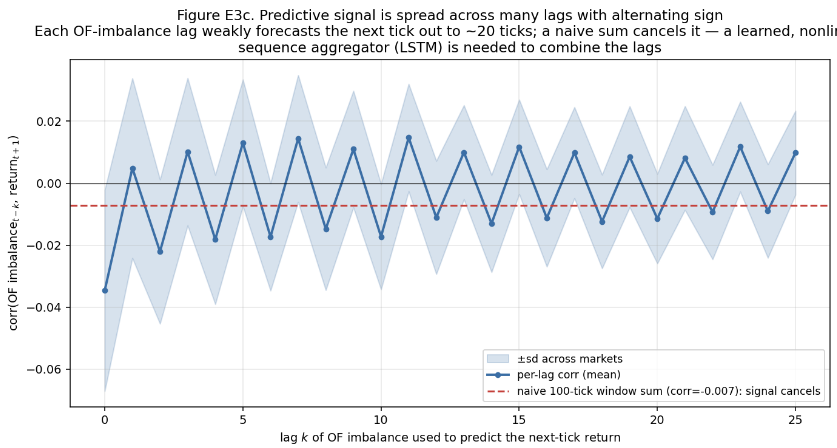

Figure 2: **Predictive signal is distributed across lags with alternating sign.** Each OFI lag weakly forecasts the next-tick return out to $\sim$20 ticks; a naive equal-weight window sum cancels these contributions, so a learned, nonlinear sequence aggregator is needed.

The target: a spike at zero with heavy tails. The mid-price return has an enormous point mass at exactly zero (85% of ticks flat at $h = 1$, falling to 69% at $h = 10$) surrounded by heavy tails on the one-cent grid (Figure 3). This justifies three choices: directional accuracy is scored only on moved ticks ($r \neq 0$), since scoring flat ticks merely rewards predicting zero; a genuine edge at tick resolution registers as an R^2_{OS} of only 10^{-3}–10^{-2}; and the ragged scale of returns is why both features and targets are standardized.

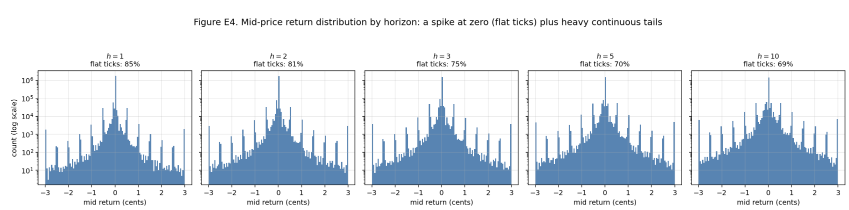

Figure 3: **Mid-price return distribution by horizon.** A dominant spike at zero (flat-tick fraction 85% at h=1 to 69% at h=10) plus heavy tails, motivating moved-tick directional accuracy, the small R^2_{OS} scale, and target standardization.

4 Models

All models share an input–output interface and differ only in the function class. The input is the 100×20 order-flow window; the output is the five-dimensional vector of return forecasts at

$h = 1, 2, 3, 5, 10$, produced jointly by one regression head; all are trained by minimizing MSE on the standardized targets (multi-horizon regression, not classification). We fit a single **pooled** model on all twelve contracts rather than per-market models: individual contracts are too short to train a recurrent network reliably, so pooling amortizes the limited data into one well-populated training set, and the standardization of Section 2 places markets on comparable scales. The cost is that the model cannot specialize to any one market (examined in Section 8). Windows are formed strictly within a single market and split, so pooling never leaks across the train/validation/test boundary.

LSTM. A single-layer LSTM (hidden size 128) consumes the 100×20 sequence one tick at a time; the final hidden state passes through a linear read-out emitting the five forecasts. There is no feature-extraction stage, so the cell learns directly from raw signed order flow. The forget-gate bias is initialized near unity (standard practice); 77,445 parameters.

LSTM-MLP. Identical recurrent trunk, but the linear read-out is replaced by a one-hidden-layer MLP (width 64, ReLU). This is a strict generalization of the linear head, isolating whether a nonlinear transform of the recurrent representation helps; 85,381 parameters.

CNN-LSTM. A convolutional stack acts as a learned local feature extractor—first across price levels, then across short tick spans—followed by a recurrent layer and a linear read-out (Table 2). Each conv block is Conv $\rightarrow$ batch-norm $\rightarrow$ LeakyReLU(0.01); the batch-norm layers, not part of the original DeepLOB specification (Zhang et al., 2019), were added to stabilize optimization. 125,125 parameters.

Table 2: CNN-LSTM architecture.

Stage	Operation	Output shape
Input	Order-flow window	$100 \times 20 \times 1$
Block 2a	Conv (1×2), stride (1×2), 32 filters (merges bid/ask)	$100 \times 10 \times 32$
Block 2b	Two Conv (4×1), time same-padding, 32 filters	$100 \times 10 \times 32$
Block 3	Conv (1×10), 32 filters (aggregates levels)	$100 \times 1 \times 32$
Inception	3-tick conv, 5-tick conv, max-pool (64 ch. each), concatenated	100×192
LSTM	One layer, hidden size 64; final state retained	64
Dense	Linear read-out, one value per horizon	5

Benchmarks. The *naive* predictor forecasts the training-mean return at every horizon; on standardized targets it attains MSE one, defining the zero line for R^2_{OS} and the 0.50 line for directional accuracy. The *Ridge* benchmark flattens the 100×20 window into a 2,000-dimensional vector and fits a ridge model ($\alpha = 1$) jointly across horizons—a linear-autoregressive proxy. Both consume the identical window and forecast the same horizons, so each comparison varies only the function class.

5 Training, Validation, and Testing

Within each market the cleaned tick sequence is cut chronologically into a validation set (earliest 15%), training set (next 70%), and test set (most recent 15%); windows never straddle a boundary. The chronological cut prevents look-ahead bias and guarantees training precedes test in time; placing training immediately before test minimizes the temporal gap given the non-stationarity, while validation sits furthest away because it is used only for selection. Every standardization statistic and the naive baseline are estimated on training and frozen for the other splits. Windows are sampled at every s-th tick (the stride); an evaluation stride of ten yields $\approx$ 30,500 validation and $\approx$ 30,600 test windows. For Section 8 the pooled model is additionally scored on each market's test segment in isolation, reusing the global standardization and naive baseline.

6 Hyperparameter Selection

Every selection decision used a composite score on the validation set; the test set played no role:

$$\text{score} = 2(\text{DirAcc} - 0.5) + \text{Sharpe} + 100\, R^2_{\text{OS}}, \tag{2}$$

which renders the three terms commensurable (directional accuracy rescaled to $\approx [0, 1]$, Sharpe already of order one, and the validation $R^2_{\text{OS}} \approx 0.01$–$0.02$ multiplied by 100). The sweep covered learning rate, weight decay, hidden size, and training stride, with grids differing slightly across architectures (the LSTM variants additionally probed smaller learning rates down to 10^{-5} and hidden sizes up to 256). Only the CNN-LSTM had its two finalists re-estimated across three seeds $(0, 7, 42)$ to account for the high Sharpe variance; the LSTM and LSTM-MLP winners are reported at seed 42. Two findings held across all three: (i) data volume was the dominant lever—reducing the training stride from 25 to 5 raises the window count roughly fivefold (from $\approx 57{,}000$ to $\approx 286{,}000$) and produced the largest, most consistent gains; and (ii) regularization conferred no benefit (zero weight decay optimal throughout), consistent with the train–test gap reflecting distributional shift rather than over-fitting. Winning configurations are in Table 3.

Table 3: **Winning hyperparameters.** Shared across models: look-back $W = 100$, evaluation stride 10, z-scored targets, Adam, batch size 256, max 60 epochs, gradient clipping (max-norm 1.0), weight decay 0, MSE on standardized targets, seed 42.

	LSTM	LSTM-MLP	CNN-LSTM
Training stride	5 (≈286k)	10 (≈143k)	5 (≈286k)
Hidden size	128	128	64
MLP head	—	64 (ReLU)	—
Conv / Inception filters	—	—	32 / 64
Learning rate	2×10^{-3}	10^{-3}	10^{-3}
Early-stop patience	8	8	6
Trainable parameters	77,445	85,381	125,125

Early stopping monitors the validation loss (the same standardized MSE minimized in training, not the composite score) and restores the best-validation weights before testing; predictions are then returned to raw units. Training ran on Modal accelerators (CNN-LSTM on an H100, the two LSTM variants on an A10G). The loss curves for the winning CNN-LSTM run (Figure 4) show a ≈ 3.5-fold train–validation gap that does not narrow with weight decay; this reflects non-stationarity between the periods rather than classical over-fitting.

7 Evaluation Metrics

All metrics are computed in raw return units. (i) The out-of-sample $R^2_{\text{OS}} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{naive}}$; positive means beating the train mean, and at tick resolution 10^{-3}–10^{-2} already signals a genuine edge while a value near 0.5 would indicate leakage. (ii) Directional accuracy $\text{DirAcc} = \text{mean}(\mathbb{1}[\hat{s} = s])$ with $\hat{s} = \text{sign}(\hat{r})$ and $s = \text{sign}(r)$, over moved ticks ($r \neq 0$), chance baseline 0.50. (iii) The annualized Sharpe ratio of a frictionless unit-position strategy, $\text{pnl}_t = \text{sign}(\hat{r}_t)\, r_t$ and $\text{Sharpe} = [\text{mean}(\text{pnl})/\text{std}(\text{pnl})]\sqrt{252}$; here $\sqrt{252}$ is a fixed scaling constant for cross-model comparability, not a literal calendar annualization of tick-level P&L, and Sharpe is the noisiest metric across seeds.

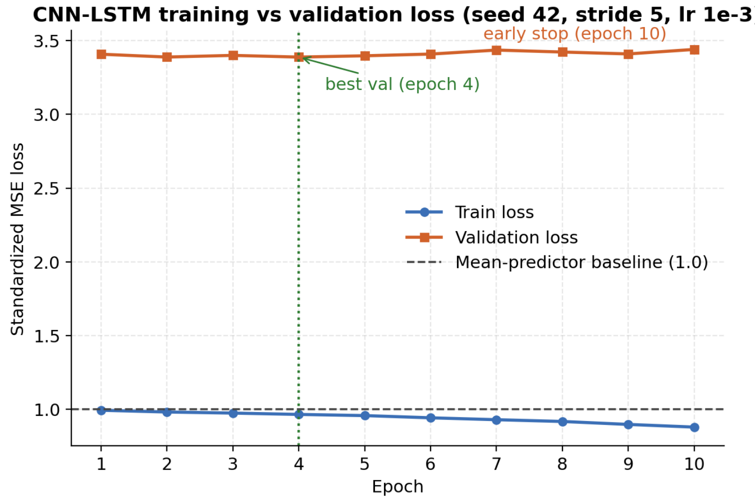

Figure 4: **CNN-LSTM training and validation loss (seed 42).** Standardized MSE by epoch, with the best-validation and early-stopping epochs marked and a dashed mean-predictor baseline at one. The $\sim$3.5-fold gap reflects non-stationarity, not over-fitting.

8 Results

The order-flow signal is real and beats both baselines. Table 4 reports in- and out-of-sample metrics. Every neural model clears both references: directional accuracy $\approx$ 0.60–0.61 on moved ticks (above the 0.50 chance level and Ridge's 0.557) and positive Sharpe of 0.39–0.55 (vs. naive 0 and Ridge 0.12). The tuned CNN-LSTM is strongest on every test metric—directional accuracy 0.6134, Sharpe 0.5514, and the only strictly positive test R^2_{OS} (+0.0007 at seed 42; seed-averaged $+0.0017 \pm 0.0007$, positive on four of five horizons). Ridge posts a *negative* R^2_{OS} (-0.0115): a linear fit over 2,000 noisy inputs is beaten by simply predicting the train mean. Economically, order-flow imbalance carries genuine information about the next few mid-moves in these contracts—the same signal documented on Nasdaq equities (Kolm et al., 2023), now in a market that prices probabilities. An $R^2_{\mathrm{OS}} \sim 10^{-3}$ is what microstructure theory predicts at tick resolution, where almost all return variance is unforecastable noise; a value near 0.5 would signal leakage. Notably, the Ridge R^2_{OS} is *more* negative in sample (-0.0484) than out (-0.0115), because the train-mean baseline is an especially strong reference on the very window from which its mean is computed.

Table 4: **In-sample vs. out-of-sample performance**, horizon-averaged at seed 42. "in" is the training split, "test" the test split. Higher is better. *CNN-LSTM seed-averaged test:* $R^2_{\mathrm{OS}} = +0.0017 \pm 0.0007$*,* $\mathrm{DirAcc} = 0.6225 \pm 0.0068$*,* $\mathrm{Sharpe} = 0.5666 \pm 0.033$*.*

	R^2_{OS}		Directional acc.		Sharpe (ann.)		
Model	in	test	in	test	in	test	Params
CNN-LSTM	0.0456	+0.0007	0.6159	0.6134	0.8518	0.5514	125k
LSTM	0.0363	−0.0005	0.6362	0.6085	0.7849	0.3851	77k
LSTM-MLP	0.0415	−0.0017	0.6234	0.5969	0.8176	0.4785	85k
Ridge linear	−0.0484	−0.0115	0.5592	0.5565	0.3618	0.1183	—
Naive (train-mean)	0.0000	0.0000	0.5000	0.5000	0.0000	0.0000	—

Architectural complexity adds little. The CNN-LSTM's margin over a plain LSTM is slight (0.6134 vs. 0.6085 directional accuracy) despite 1.6× more parameters and a full convolutional stack, and the LSTM-MLP's nonlinear head buys nothing (0.5969). This reproduces the headline of Kolm et al. (2023): once informative order-flow features are supplied, a simple recurrent model captures most of the signal. The value resides in the feature representation—signed, level-resolved order flow—not in depth, so the cheaper model is the sensible production choice.

Predictability is concentrated at the nearest tick. For the CNN-LSTM, directional accuracy is highest at $h = 1$ (0.696) and decays to 0.562 at $h = 10$ (Figure 5); the LSTM (0.709 → 0.531) and LSTM-MLP (0.706 → 0.535) show the same shape. The immediate next move is by far the most forecastable from recent order flow—the prediction-market echo of the equity finding that microstructure alpha decays after a couple of price changes (Kolm et al., 2023; Zhang et al., 2019). Figures 5–7 break each model down by horizon; the three bars per horizon are in-sample, test (seed 42), and the out-of-sample Ridge benchmark, on common axes. Because the edge is concentrated at $h = 1$ and fades within a handful of ticks, it must be harvested almost immediately: any latency or holding period beyond a few ticks erodes the confident horizons.

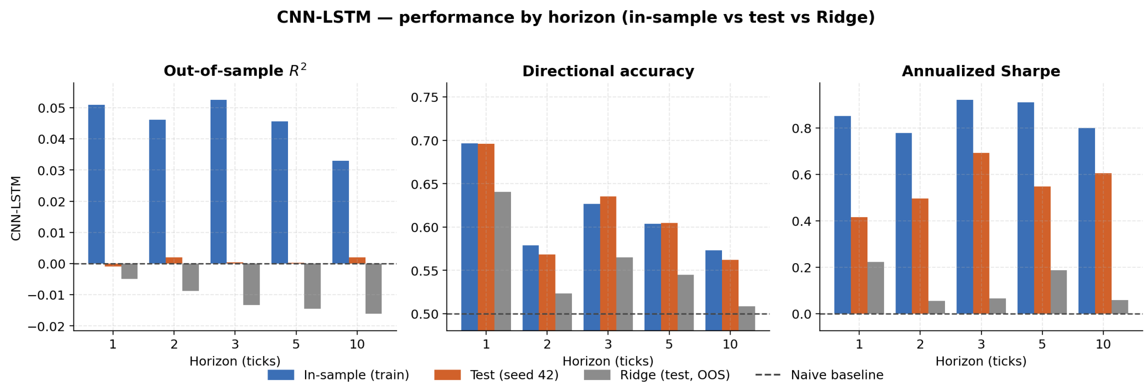

Figure 5: **CNN-LSTM performance by horizon.** Out-of-sample R^2 (left), directional accuracy (center), and annualized Sharpe (right). The in-sample-to-test gap is pronounced for R^2 but modest for directional accuracy and Sharpe; the CNN-LSTM sustains the highest test bars at the near horizons and is the only model with a positive out-of-sample R^2.

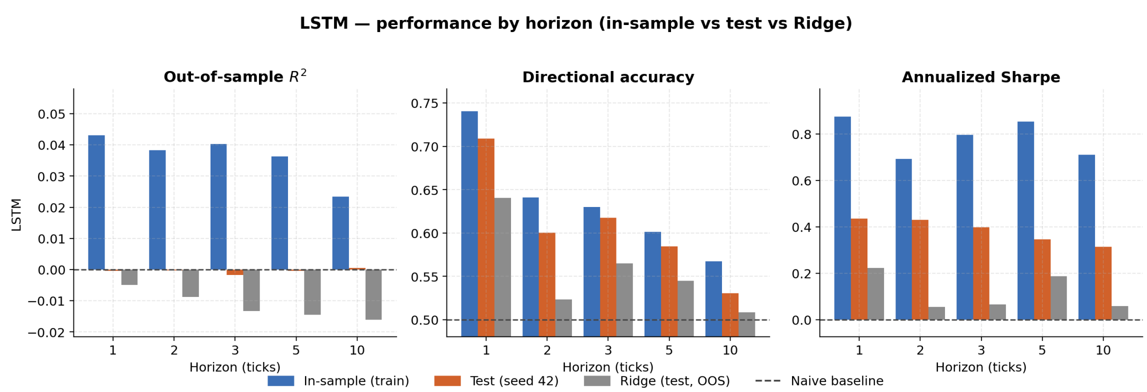

Figure 6: **LSTM performance by horizon.** The plain single-layer LSTM nearly matches the CNN-LSTM on directional accuracy at 1.6× fewer parameters, with the same near-tick concentration of skill.

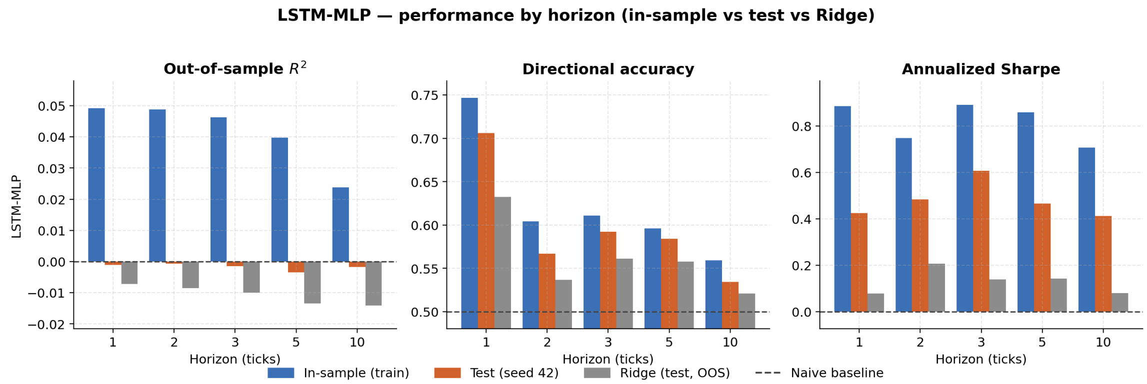

Figure 7: **LSTM-MLP performance by horizon.** Adding a nonlinear MLP head to the recurrent trunk leaves performance essentially unchanged relative to the plain LSTM, confirming that the gain comes from the order-flow representation rather than head complexity.

The squared-error edge is fragile, and the Sharpe is frictionless. The validation R^2_{OS} for the LSTM and LSTM-MLP ($+0.015$, $+0.012$) collapses to essentially zero on test (-0.0005, -0.0017); even the CNN-LSTM falls from $+0.0159$ to $+0.0007$. This is not classical over-fitting (weight decay never helped, training loss falls smoothly, directional accuracy holds up) but non-stationarity: as a best-of-seven series unfolds, information arrives in discrete jumps (results, injuries, rotations) and the probability path's dynamics shift, so a model fit on earlier ticks faces a different test distribution. The recoverable edge is therefore *directional, not magnitude*: the sign generalizes while the calibrated size does not, so a deployable strategy should size by predicted sign. Second, the Sharpe ratios are *frictionless*—a unit position every tick with zero trading cost—so the $\sim$0.55 figure is evidence of predictive information, not a profitable rule: Polymarket spreads (≈ 0.007–0.020) dwarf the per-tick return, and a tick-flipping rule would not survive fees and slippage. We treat these figures as an upper bound on deployable performance.

Robustness to volatility. We test whether the pooled model's performance depends on a contract's price-curve volatility (H_0: independent of volatility). Grouping the twelve markets into Low/Mid/High realized-volatility tertiles and scoring each market's test segment (Table 5, Figure 8), directional accuracy stays in roughly 0.60–0.65 and the Sharpe ratio remains positive in every group, so the model's directional skill does not vary systematically with the price curve. The out-of-sample R^2 is negative in every group, but this is a metric artifact: R^2_{OS} is relative to each market's own train-mean baseline, which a short, quiet test segment makes especially hard to beat. Spearman correlations across the twelve markets confirm this: none of the three metrics correlates significantly with full-life volatility (directional accuracy $\rho = 0.06$, $p = 0.86$; Sharpe $\rho = -0.02$, $p = 0.95$; R^2_{OS} $\rho = 0.09$, $p = 0.78$), and the only significant relationship is mechanical—R^2_{OS} rises with the volatility of the test window itself ($\rho = 0.78$, $p = 0.003$).

9 Limitations and Conclusion

Four limitations bound our conclusions. **Pooled, not per-instrument:** where Kolm et al. (2023) fit a model per instrument, our short contracts force a single pooled model that cannot specialize; Section 8 suggests its directional skill is stable across markets, but this remains a compromise. **Narrow scope:** twelve NBA series-winner contracts over one playoff window is a narrow slice, and sports markets (fixed resolution date, game-by-game information jumps) may not generalize to politics, macro, or crypto venues where news arrives continuously. **Maturity is ignored:** because

Table 5: **CNN-LSTM test metrics by volatility tertile** (seed 42, four markets per group). "(w)" is the window-weighted mean. Group Sharpe figures average per-market values and so run higher than the single pooled-test Sharpe of 0.55.

Group	R^2_{OS}	DirAcc	Sharpe	R^2_{OS} (w)	DirAcc (w)	Sharpe (w)
Low (n=4)	−0.292	0.613	1.279	−0.190	0.601	0.946
Mid (n=4)	−0.147	0.646	1.331	−0.043	0.653	1.383
High (n=4)	−0.095	0.599	0.736	−0.073	0.607	0.771

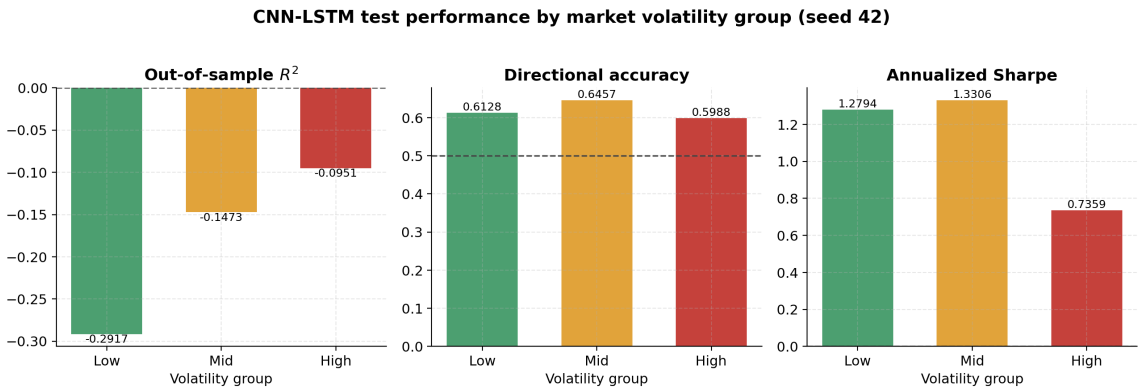

Figure 8: **CNN-LSTM test performance by volatility tertile (seed** 42**).** Directional accuracy (center) is flat across groups; Sharpe (right) dips for the High group; per-market R^2_{OS} (left) is negative in every group—a baseline artifact on short, quiet test segments, not lost skill.

the splits are chronological, the test data are always nearest the payout date and more volatile (the standardized loss gap—training ≈ 0.95 vs. validation and test ≈ 3.4–3.5—is the symptom), reflecting a likely shift in the data-generating process as a contract moves from distant-and-uncertain to near-determined; a maturity-aware model (adding time-to-resolution, or fitting by maturity bucket) could address this. **Frictionless P&L:** the Sharpe takes a unit position every tick at zero cost, so a genuine evaluation needs a cost-aware backtest with passive execution on the highest-confidence ticks.

Conclusion. Deep order-flow forecasting transfers to prediction markets—a qualified yes, and to our knowledge the first such application to a prediction-market venue. All three neural models beat both baselines on every metric (≈ 61% directional accuracy and a genuinely positive R^2_{OS} for the tuned CNN-LSTM), and both structural findings reproduce: a plain LSTM nearly matches the CNN-LSTM (the value is in the order-flow representation, not depth), and predictability is concentrated at the nearest tick. But the transfer is established at the level of *signal*, not realized profit: the squared-error edge is fragile out of sample under the non-stationarity of contracts marching toward resolution, and the economic metrics are frictionless. The robust, generalizing component is directional rather than magnitude-based. Mature equity-microstructure methods carry real information into this growing but understudied venue; turning it into net economic value requires the maturity-aware modeling and cost-aware evaluation we leave to future work.

Acknowledgments

Order-book data are from the Telonex Polymarket Historical Data API (`book_snapshot_25`). The modeling adapts Kolm et al. (2023) and DeepLOB (Zhang et al., 2019); the order-flow construction follows Cont et al. (2014). The CNN-LSTM framework is adapted from the original DeepLOB implementation, available at https://github.com/zcakhaa/DeepLOB-Deep-Convolutional-Neural-Networks-for-Limit-Order-Books. The pipeline uses Python (NumPy, pandas, PyArrow, scikit-learn, PyTorch, SciPy, Matplotlib); training ran on Modal (H100 and A10G GPUs). We used AI language-model assistants for code development, debugging, commenting, and formatting, and for editing and proofreading the report; we reviewed and verified all code and text ourselves.

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